Skyler AI vs. Existing Research — Gaps and Outcomes

The papers

	Mukherjee & Jiang (2019, arXiv)	**Zhang et al. (2021, ACL — EmailSum)**	**Prexawanprasut (2025, Thailand Statistician)**
Model	TF-IDF/GBDT + CNN-MLP ensemble, with user/email similarity features	T5 / BART abstractive summarization	BERT, DistilBERT, XLM-R
Task	Predict whether a user will act on (reply to) an incoming email	Abstractive summarization of email threads	Classify business email into 4 operational workflow categories
Reported metrics	**0.788 AUROC** (best ensemble) vs. 0.757 best baseline, on the public Avocado email dataset	No single trusted automatic metric — authors show ROUGE/BERTScore correlate weakly with human judgment on this task	**XLM-R > 92% accuracy** on 12,500 real business emails across 4 domains
What they measure	Offline triage-action ranking quality, on one fixed public dataset	Summary quality via human + automatic evaluation, fixed dataset	Offline classification accuracy, fixed labeled dataset
What they don't measure	Request-time serving cost; a *combined* multi-signal decision; privacy of the per-user email history used as model input	Whether the model is usable live inside a product that actually sends replies; serving cost; privacy	Request-time serving cost under load; a combined sentiment+keyword+semantic decision; per-decision explainability; privacy/data locality

All three are freely downloadable (arXiv / ACL anthology / Thailand's open-access TCI-Thaijo platform) — no IEEE paywall or institutional login needed for any of them.

Direct download links:

- Mukherjee & Jiang (2019): https://arxiv.org/abs/1905.01991

- Zhang et al. (2021, EmailSum): https://arxiv.org/abs/2107.14691

- Prexawanprasut (2025): https://ph02.tci-thaijo.org/index.php/thaistat/article/view/266519/174074

The gap

Each paper validates exactly one piece of what a real inbox-management product does — deciding what to do with an email, summarizing it, or classifying it — in isolation, offline, on a fixed dataset. None of the three measures serving-time cost on a live backend, whether a *combined* decision built from several signals together (not one model's output) is correct, or where user data goes during inference — despite data locality being a natural concern for any tool that reads a user's inbox.

Gap-by-gap: what each paper doesn't do, and what Skyler AI does instead

vs. Mukherjee & Jiang (2019) — triage-action prediction

Their gap: Predicts one fixed thing — will the user reply? — learned from a user's past email history, on one offline dataset (Avocado). The category of "action" is fixed by the researchers; it's not something the end user can configure, and it isn't wired into any live product that actually does something with the prediction.

What Skyler AI addresses: Skyler AI gives the *user* multiple independent inboxes, each with its own rule the user designs themselves — sentiment, keyword, and semantic-similarity conditions combined however they choose — not a fixed category like "spam" or "starred" or "will I reply." The decision is live, not an offline prediction: it runs in production, returns results in one click, and is now measured end-to-end at 94.4% accuracy on the actual deployed combined decision (Section "End-to-end decision accuracy" below), not just a ranking score on a static dataset.

vs. Zhang et al. (2021) — EmailSum (summarization)

Their gap: Studies whether a model can summarize an email thread well, evaluated offline against human judgments. They explicitly found automatic metrics unreliable for this and stopped short of deploying it — there's no live system, no reply-sending action, nothing automated end-to-end.

What Skyler AI addresses: Skyler AI's summarization and reply-drafting is live and automated: once a user sets a rule for an inbox, matched emails can be summarized and a reply drafted automatically, then sent to Gmail in one click — not a research benchmark, an actual automation a user runs repeatedly without re-configuring anything.

vs. Prexawanprasut (2025) — workflow classification

Their gap: Classifies business email into four *fixed* categories chosen by the researchers, measured once offline on a static 12,500-email set. Even their "real-time" prototype reports no serving-cost, privacy, or combined-decision numbers.

What Skyler AI addresses: Skyler AI's categories aren't fixed by a developer — a user designs their own filter per inbox, and can run as many independent inboxes as they need, each organized differently. On top of that, Skyler AI proves (not claims) what Prexawanprasut's prototype never measures: request-time serving cost (252–310 ms bug found and fixed), zero network egress during classification (proven by a live test), and combined multi-signal decision accuracy (94.4%).

What none of the three papers address at all

- **Local-only inference, not cloud-dependent.** All three papers' models could in principle be run anywhere; none of them measure or prove where the data actually goes. Skyler AI's classification (sentiment + semantic matching) runs entirely on local, open-weight models (DistilBERT, MiniLM) in-process — proven via `verifyNoNetworkEgress.ts` to make zero network calls. The only cloud call in the whole product is the *optional*, user-initiated reply-draft step (Groq), never the classification/filtering decision itself.

- **No per-classification API cost.** Because classification runs locally, it costs nothing per email to classify — unlike a cloud-LLM-based classifier, which pays an API fee per call. (Skyler AI's credits system limits how many emails a user *pulls* per month on the free tier; it is not a per-classification AI fee.)

- **Local caching.** Pulled emails are cached client-side (Dexie/IndexedDB) rather than held server-side indefinitely — another privacy-relevant design choice none of the three papers' systems address, since none of them are deployed, multi-user products with a client at all.

Skyler AI's outcome, measured against that same gap

Axis	What the papers report	Skyler AI, measured
Classification quality	0.788 AUROC (triage-action ranking); no single trusted metric (summarization); >92% accuracy (4-category workflow classification)	**89.91% accuracy / 90.27% F1** on the real, official SST-2 sentiment benchmark — a different task from all three, stated plainly, not claimed as a win

Axis	What the papers report	Skyler AI, measured
Request-time serving cost	Not measured by any of the three	**252–310 ms per request** of model-reload overhead identified and eliminated; live `classifyMs` telemetry on the real endpoint
Per-email classification cost	Not measured by any of the three	**~22–26 ms/email** at realistic 200-email scale, no LLM call per email
End-to-end decision accuracy (sentiment AND keyword/semantic match combined)	Not measured by any of the three	**94.4% accuracy, 100% precision, 83.3% recall, 90.9% F1** — measured on a labeled 90-email test set against the real, deployed system
Explainability	None of the three expose a reason for the combined decision a product would actually act on	**Every kept/discarded email now returns why**: sentiment label, matched keyword (if any), semantic-match flag — for the actual combined decision, live on the endpoint (`filterEmailsWithReasons`)
Privacy / data locality	Not measured or proven by any of the three	**Proven, not claimed**: a network-call interceptor patched into Node's http/https/net/tls/fetch before model load records zero network calls across model loading, sentiment analysis, and semantic embedding — verifiable by anyone who runs `verifyNoNetworkEgress.ts`
Latency predictability under real load	Not measured by any of the three	**1.2x–8.2x** reduction in latency variance; worst-case spikes bounded from 11–14 sec down to ~8 sec

Bottom line

Skyler AI is a multi-inbox organizer: it filters, prioritizes, and summarizes/drafts replies for Gmail using local models, not a spam or phishing filter. The closest literature match on raw classification accuracy — Prexawanprasut's 92%+ business-email workflow classifier — targets a different label (workflow category vs. keep/discard), so no single-number "we beat them" claim is made; that's stated plainly. On every other axis — request-time serving cost, per-email cost, combined-decision accuracy, explainability of that combined decision, and privacy — **none of the three papers reports a number or a proof at all**. Skyler AI does, against a real, deployed system, with the method open for anyone to re-run.